

Technical Summary: Cranial Index & Cranial Vault Asymmetry Index Calculator Pro

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Directory Location: /cranial_index/

Part 1: Clinical Indices & Mathematical Formulas

1. Cranial Index (CI) - Proportions (Brachycephaly / Scaphocephaly)

Calculated by dividing maximal Biparietal Diameter (BPD) by Occipitofrontal Diameter (OFD):

$$\text{CI} = \frac{\text{Biparietal Diameter (BPD)}}{\text{Occipitofrontal Diameter (OFD)}}$$

- **Scaphocephaly / Dolichocephaly:** $\text{CI} < 0.75$ (< 75.00%)
- **Mesocephaly (Normal):** $0.75 \leq \text{CI} \leq 0.85$ (75.00% – 85.00%)
- **Brachycephaly:** $\text{CI} > 0.85$ (> 85.00%)
- **Severe Brachycephaly:** $\text{CI} > 0.90$ (> 90.00%)

2. Cranial Vault Asymmetry Index (CVAI) - Dynamic Denominator Option

Calculated from **Shorter diagonal diameter** and **Longer diagonal diameter** taken at 30° to the midline sagittal axis:

$$\text{CVAI (\%)} = \frac{|\text{Longer Diagonal} - \text{Shorter Diagonal}|}{\text{Chosen Denominator Diagonal}} \times 100$$

- **Standard Denominator (Shorter Diagonal):** Default convention used for standard clinical threshold bands.
- **Alternative Denominator (Longer Diagonal):** Available via real-time user selection dropdown.
- **Normal Symmetry:** $\text{CVAI} < 3.5\%$
- **Mild Plagiocephaly:** $3.5\% \leq \text{CVAI} < 6.25\%$
- **Moderate Plagiocephaly:** $6.25\% \leq \text{CVAI} \leq 8.75\%$
- **Severe Plagiocephaly:** $\text{CVAI} > 8.75\%$
- **CVA (Unnormalised Difference):** $\text{CVA} = |\text{Longer} - \text{Shorter}|$ (in mm/cm)

Denominator Literature Reference & Finding:

Regardless of whether the *shorter* or *longer* diagonal diameter is used as the denominator, research demonstrates little practical clinical difference in threshold values.

Miyabayashi H, Saito K, Kato R, Noto T, Nagano N, Morioka I. Denominator of Cranial Vault Asymmetry Index: Choosing Between Longer and Shorter Diagonal Lengths. *J Craniofac Surg.* 2023;33(5):e369. doi:10.1097/SCS.00000000000009263

Mandatory Clinical Threshold Note: The conventional cut-off thresholds above were derived using the shorter diagonal diameter as the denominator. Diagnosis is clinical; values can vary depending on age, sex, and race/ethnicity.

3. WHO Head Circumference Centiles & Growth Chart (0 to 60 Months)

Calculates exact Z-scores and centiles for pediatric patients based on the **WHO Child Growth Standards** (Birth to 5 Years / 60 Months):

$$Z = \frac{\text{Head Circumference (cm)} - M(\text{age, gender})}{SD(\text{age, gender})}$$

- **Gender Standards:** Separate WHO growth curve datasets for Boys and Girls.
- **Percentile Calculation:** Exact standard normal cumulative distribution function ($\Phi(Z)$) outputting centiles from 0.1% to 99.9%.
- **Macrocephaly:** Head Circumference > 97th Centile (+2.0 SD)
- **Normal Vault Range:** 3rd Centile ≤ Head Circumference ≤ 97th Centile (-2.0 SD to +2.0 SD)
- **Microcephaly:** Head Circumference < 3rd Centile (-2.0 SD)

Growth Standard Reference:

World Health Organization. *WHO Child Growth Standards: Head circumference-for-age, 0 to 60 completed months.* Geneva: World Health Organization; 2006.

⚠ Part 2: Critical Clinical Points

- **Scaphocephaly Reassurance Alert:** A normal CVAI does *not* reassure in scaphocephaly. Sagittal synostosis is a symmetric deformity, so CVAI remains normal while CI falls.
- **Morphology vs. Metric:** CVAI cannot distinguish deformational plagiocephaly (parallelogram deformation) from synostotic plagiocephaly / unicoronal / lambdoid synostosis (trapezoid deformation). Morphological clinical examination is required.

📁 Repository Files Summary

File Name	Description
<code>index.html</code>	Structural HTML5 layout, Pro title, dual-tab bar, Head Circumference (cm) & Date picker textboxes, relabeled Shorter/Longer diagonal columns, denominator dropdown, and Miyabayashi et al. reference citation.
<code>style.css</code>	Glassmorphism dark aesthetic, denominator selector box styling, literature citation formatting, and active-tab <code>@media print</code> print rules.

File Name	Description
script.js	Dual-tab state manager, dynamic denominator CVAI calculation engine, CounterAPI v2 global counter integration, and Chart.js bar renderers.
cranial_diagram.jpg	Schematic vertex view showing glabella, opisthocranium, BPD, and OFD diameters.
cvai_diagram.png	Updated schematic vertex view showing 30° obliques labeled A (Shorter) and B (Longer).

Please check the calculations manually. Disclaimer: This tool is not intended for clinical decision support; it is for educational reference in patients with craniosynostosis and cranial deformities. Final diagnosis and treatment decisions require correlation with comprehensive clinical examination and multiplanar radiological/anthropometric assessment. Threshold values provided can vary depending on the age, sex, and race/ethnicity of the child.